

RDK8 Firebolt® JSON-RPC Specification

Document status	APPROVED
Author	Andrew Bennett
Reviewers	Bart Catrysse Tomasz Blaszczak Vladimir Rangelov Sathishkumar Deena Kirupakaram

Summary

This document defines the JSON-RPC messages that are exchanged between Firebolt apps/runtimes and the Firebolt API Gateway.

Related Documents

- [JSON-RPC 2.0 Specification](#)

Definitions

Name	Meaning
Primitive type	NULL, Boolean, Number or String

JSON-RPC 2.0 Compliance

- The URL used by the Firebolt Client Library to connect to the Firebolt API Gateway includes a query parameter that states the the Client Library will use and accept only compliant JSON-RPC as specified in this document - the query parameter is: `RPCv2=true`
- All names that are used for matching, such as the names of methods and parameters, are case sensitive, as stated in Section 2 of the JSON-RPC 2.0 Specification

Method Calls

	Message	Example	Notes
1	Method call with no parameters	<pre>{ "jsonrpc": "2.0", "id": 1, "method": "Module.method" }</pre>	Params is not provided
2	Method call with one or more parameters	<pre>{ "jsonrpc": "2.0", "id": 1, "method": "Module.method", "params": { "attr1": 1, "attr2": "value2" } }</pre>	Params is an object. Arrays are not used
3	Error response	<pre>{ "jsonrpc": "2.0", "id": 1, "error": { "code": -40300, "message": "Method not permitted" } }</pre>	

4	Success with no return value	{ "jsonrpc": "2.0", "id": 1, "result": null }	
5	Success returning a single primitive value	{ "jsonrpc": "2.0", "id": 1, "result": 42 }	result is a primitive type
6	Success returning a single attribute that is an array of values of the same primitive type	{ "jsonrpc": "2.0", "id": 1, "result": ["spa", "eng"] }	
7	Success returning more than one value	{ "jsonrpc": "2.0", "id": 1, "result": { "attr1": 1, "attr2": "value2" } }	

Notifications

	Message	Example	Notes
1	Subscribe for notification	{ "jsonrpc": "2.0", "id": 2, "method": "Module. onSomething", "params": { "listen": true } }	
2	Unsubscribe from notification	{ "jsonrpc": "2.0", "id": 2, "method": "Module. onSomething", "params": { "listen": false } }	
3	Subscription request failed	{ "jsonrpc": "2.0", "id": 2, "error": { "code": -12345, "message": "Subscription request failed" } }	
4	Subscription request succeeded	{ "jsonrpc": "2.0", "id": 2, "result": null }	
5	Notification of a primitive type	{ "jsonrpc": "2.0", "method": "Module. onSomething", "params": { "value": 42 } }	The params object has a single member that is always called "value"

6	Notification of a single attribute that is an array of values of the same primitive type	<pre>{ "jsonrpc": "2.0", "method": "Module. onSomething", "params": ["spa", "eng"] }</pre>	Params is an array that represents one attribute, not positional parameters in this case C++ example: void onSomething(std::vector<std::string> bar)
7	Notification of an object	<pre>{ "jsonrpc": "2.0", "method": "Module. onSomething", "params": { "attr1": 1, "attr2": "value2" } }</pre>	params contains all the attributes of the object
8	Notification of a list of objects	<pre>{ "jsonrpc": "2.0", "method": "Module. onSomething", "params": [{ "attr1": 1, "attr2": "value2" }, { "attr1": 2, "attr2": "value4" }] }</pre>	Params is an array that represents one attribute, not positional parameters in this case. Rarely used